Quantum Computing

Lecture 1: Introduction and Quantum Theory

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Quantum Computing

Professor Mark Oskin (University of Washington, CS&E)

1. Sunk his boat.
2. Salvaged (recovered) his boat.
3. Dropped his boat on his foot.
4. Bad for Mark, but good for me!

Slides also available at

http://www.cs.washington.edu/homes/dabacon/teaching/siena

Warning!

Two foreign languages for the price of one speaker:

English and Quantum Theory

Do not be shy!

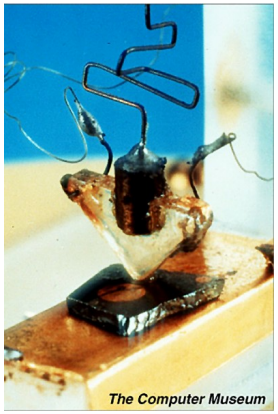
Please, ask me to repeat, to rephrase, or to just plain slow down.

In the Beginning…



Alan Turing

1936- Alan Turing: “On computable numbers, with an application to the Entscheidungsproblem”



1947- First transistor



1958- First integrated circuit

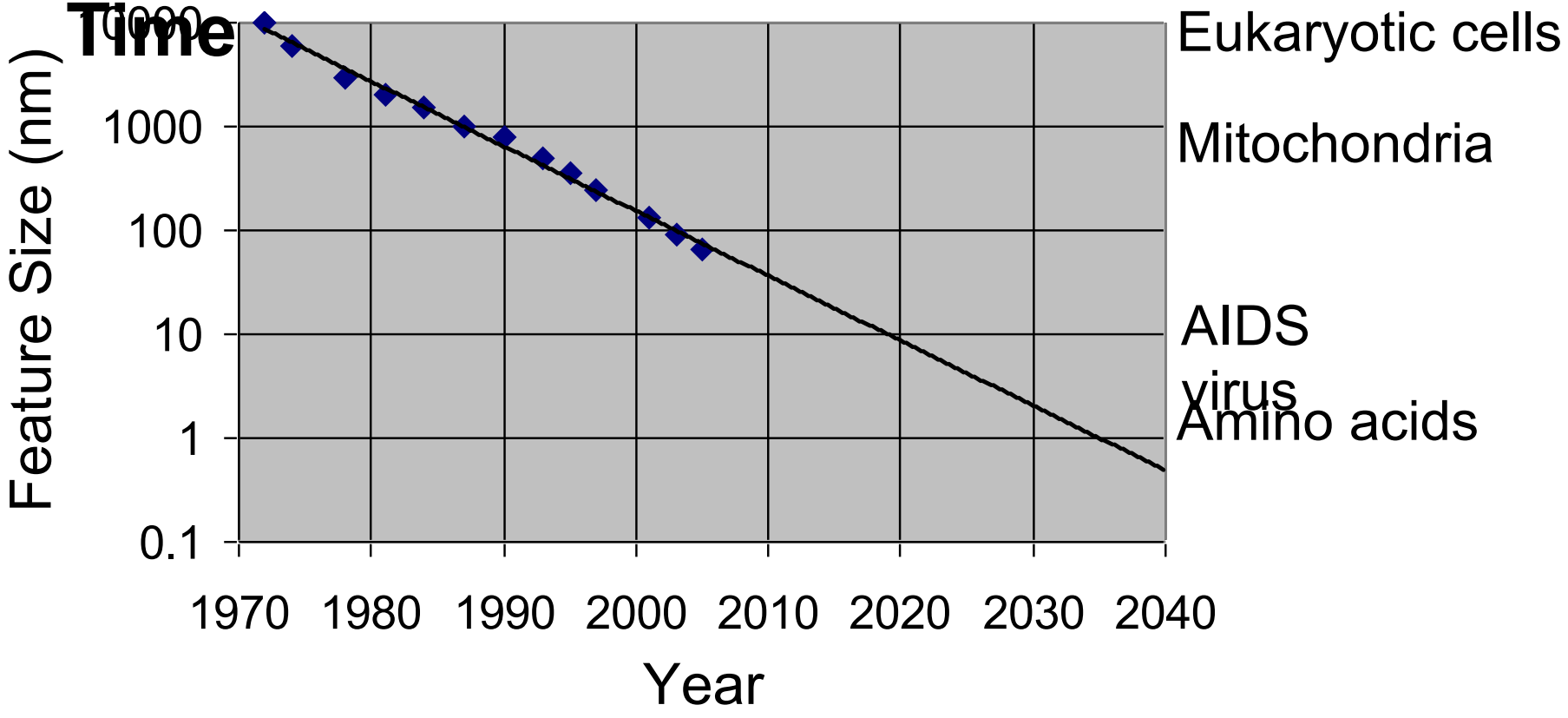


1975- Altair 8800



2005 GHz machines that weight ~ 1 pound

Moore's Law


Computer Chip Feature Size versus Time
Feature Size (nm)
10000
1000
100
10
1
0.1
1970
1980
1990
2000
2010
2020
2030
2040
Year
Eukaryotic cells
Mitochondria
AIDS virus
Amino acids

This Is the End?

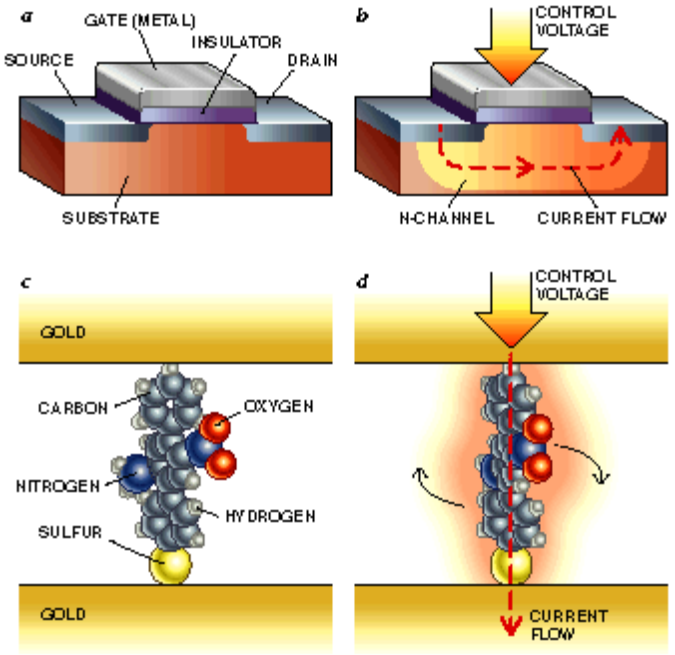
1. Ride the wave to atomic size computers?



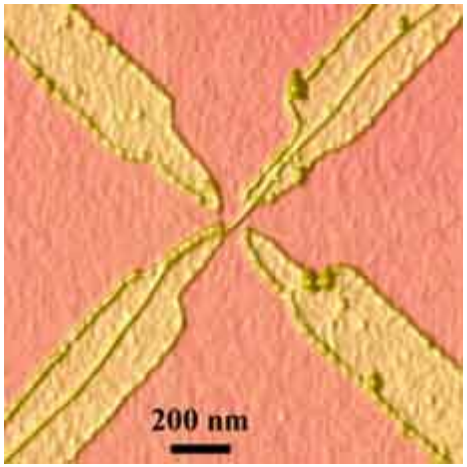
2. How do machines of atomic size operate?

Argument by Unproven Technology

1. Ride the wave to atomic size computers?



molecular transistors



carbon nanotube transistors

Pictures: http://www.mtmi.vu.lt/pfk/funkc_dariniai/nanostructures/molec_computer.htm
http://www.cemes.fr/r2_rech/r2_sr2_gns/th1_3_3_2_combing.htm

This Is the End?

2. How do machines of atomic size operate?

“Quantum Laws”

“Classical Laws”



“Quantum Computers?”

This Is the End?

2. How do machines of atomic size operate?



Richard Feynman

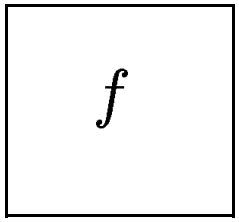


David Deutsch



Paul Benioff

Query Complexity



$$f: \quad \{0,1\}^n \quad \rightarrow \quad \mathcal{S}$$

n bit strings set

$$P(f): f \rightarrow \mathcal{T}$$

set of properties

How many times do we need to query f in order to determine $P(f)$?

Example $f: \{0,1\}^n \rightarrow \{0,1\}$:

$$P(f) = \begin{cases} 0 & \text{if } f(x) = 0, \forall x \\ 1 & \text{if otherwise} \end{cases}$$

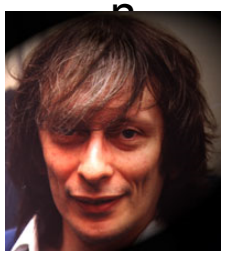
Promise problem: restricted set of functions domain of P not all f

The Work of Crazies



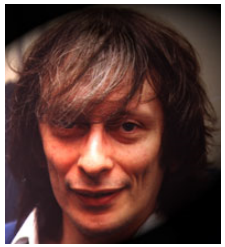
Richard Feynman

"Can Quantum Systems be Probabilistically Simulated by a Classical Computer?"



David Deutsch

1985: two classical queries
one quantum query



David Deutsch



Richard Jozsa

1992: $O(2^n)$ classical queries

2 quantum queries

$O(\log(p^{-1}))$classical queries to solve with probability of failure p

Crazies…Still Working



Umesh Vazirani



Ethan Bernstein

1993 : superpolynomially more classical than quantum queries



Dan Simon

1994 : exponentially more classical than quantum queries

The Factoring Firestorm



Peter Shor 1994

188198812920607963383697239461650439807163563379417382700763356422988859715234665485319060606504743045317388011303396716199623212057340318795506569622130516875930765025705 9

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398075086424064937397125500550386491199064362342526708406385189575946388957261768583317

×

472772146107435302536223071973048224632914695302097116459852171130520711256363590397527

Best classical algorithm takes time $O(\exp(n^{1/3}))$

Shor's quantum algorithm takes time $O(n^2 \log n)$

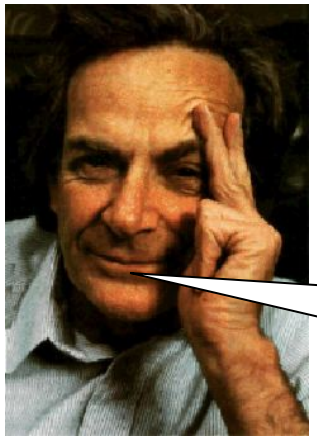
An efficient algorithm for factoring **breaks** the RSA public key cryptosystem

These Lectures

1. Quantum theory the easy way ☺ [lecture 1,2]
2. Quantum computers (circuits, teleportation) [lecture 2,3]
3. Quantum algorithms (Shor, Grover) [lecture 3,4,5]
4. Physical implementations of quantum computers [lecture 6]
5. Quantum error correction [lecture 6]
6. Quantum cryptography [lecture 6]
7. Quantum entanglement [possibly lecture 6]

Quantum Theory "The Easy Way"

Slander



I think I can safely say that nobody understands quantum mechanics.

Richard Feynman
Nobel Prize 1965

Anyone who is not shocked by quantum theory has not understood it.



Niels Bohr
Nobel Prize 1922

Quantum Th



Electromagnetis
m

Strong force

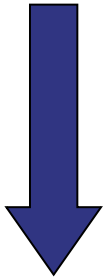
Weak force

Gravity (?)

“Quantum theory is the machine language of the universe”

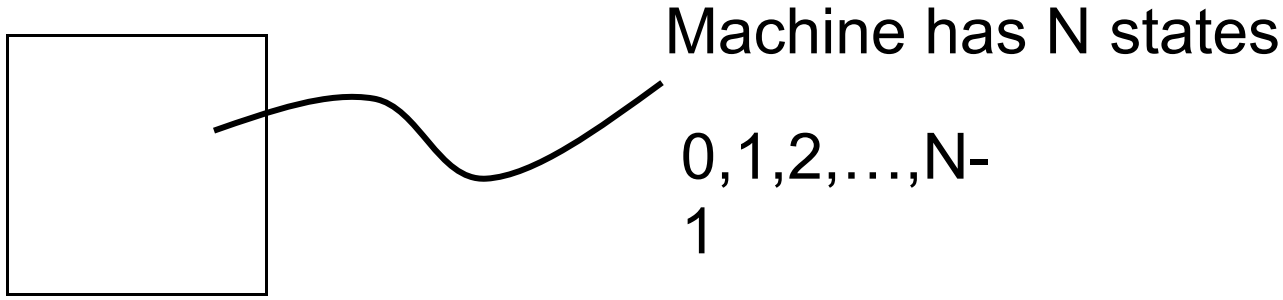
Our Path

Probabilistic classical information processing device



Quantum information processing device

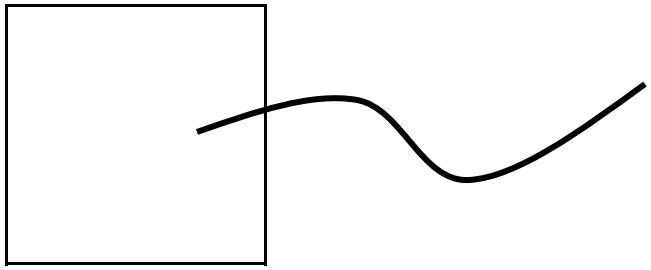
Probabilistic Information Processing Device



Rule 1 (State Description)

A probabilistic information processing machine is a machine with a state labeled from a finite alphabet of size N. Our description of the state of this system is a N dimensional real vector with positive components which sum to unity.

Rule 1



Machine has N states

0,1,2,…,N-1

N dimensional real vector

$$p \in \mathbb{R}^N \qquad p = \begin{bmatrix} p_0 \\ p_1 \\ \vdots \\ p_{N-1} \end{bmatrix}$$

positive elements

$$p_i \geq 0$$

which sum to unity

$$\sum_{i=0}^{N-1} p_i = 1$$

probability vector

Example: 3 state device

$$p = \begin{bmatrix} 0.3 \\ 0.7 \\ 0 \end{bmatrix}$$

30 % state 0

70 % state 1

0 % state 2

Probabilistic Information Processing Device

Rule 1 (State Description) N states, probability vector

Rule 2 (Evolution)

The evolution in time of our description of the device is specified by an N x N stochastic matrix A, such that if the description of the state before the evolution is given by the probability vector p then the description of the system after this evolution is given by q=Ap.

Rule 2

Evolution:

If we are in state 0, then with probability Aj,0 switch to state j

If we are in state 1, then with probability Aj,1 switch to state j

⋮

If we are in state N, then with probability Aj,N switch to state j

N2 numbers

probability to be in state j after evolution $q_j = \sum_{i=0}^{N-1} A_{j,i} p_i$

$$\begin{bmatrix} q_0 \\ \vdots \\ q_{N-1} \end{bmatrix} = \begin{bmatrix} A_{0,0} & \cdots & A_{0,N-1} \\ \vdots & & \vdots \\ A_{N-1,0} & \cdots & A_{N-1,N-1} \end{bmatrix} \begin{bmatrix} p_0 \\ \vdots \\ p_{N-1} \end{bmatrix}$$

$$q = Ap$$

Rule 2

$$q_j = \sum_{i=0}^{N-1} A_{j,i} p_i$$

these are probabilities

$$A_{j,i} \geq 0$$

$$\sum_{j=0}^{N-1} A_{j,i} = 1$$

stochastic matrix

If in state 0 switch to state 0 with probability 0.4

If in state 0 switch to state 1 with probability 0.6

If in state 1 always stay in state 1

$$A = \begin{bmatrix} 0.4 & 0 \\ 0.6 & 1 \end{bmatrix} \quad p = \begin{bmatrix} 0.4 \\ 0.6 \end{bmatrix} \qquad \begin{bmatrix} 0.4 \\ 0.6 \end{bmatrix} \rightarrow \begin{bmatrix} 0.16 \\ 0.82 \end{bmatrix}$$

$$q = Ap = \begin{bmatrix} 0.4 & 0 \\ 0.6 & 1 \end{bmatrix} \begin{bmatrix} 0.4 \\ 0.6 \end{bmatrix} = \begin{bmatrix} 0.16 \\ 0.84 \end{bmatrix}$$

Probabilistic Information Processing Device

Rule 1 (State Description) N states, probability vector

Rule 2 (Evolution) N x N stochastic matrix

Rule 3 (Measurement)

A measurement in the computational basis of our N dimensional system which is described by a probability vector p yields outcome j given by the jth component of p. After the measurement, the system is described by a probability vector with certainty that the state is j.

Rule 3

If we simply look at our device, then we see the states with the probabilities given by the probability vector.

$$p = \begin{bmatrix} p_0 \\ p_1 \\ \vdots \\ p_{N-1} \end{bmatrix}$$

If we observe the state j, then the new probability of our system is given by a probability vector with a single nonzero in the jth component:

$$p = \begin{bmatrix} p_0 \\ p_1 \\ \vdots \\ p_{N-1} \end{bmatrix} \longrightarrow \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \leftarrow \text{j}$$

Probabilistic Information Processing Device

Rule 1 (State Description) N states, probability vector

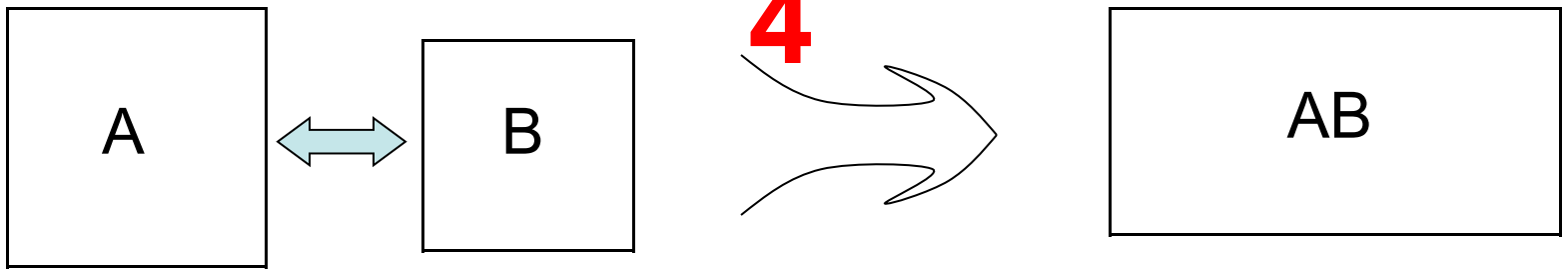
Rule 2 (Evolution) N x N stochastic matrix

Rule 3 (Measurement) k conditional probability vectors

Rule 4 (Composite Systems)

Two devices can be combined to form a bigger device. If these devices have N and M states, respectively, then the composite system has NM states. The probability vector for this new machine is a real NM dimensional probability vector from $\mathbb{R}^N \times \mathbb{R}^M$.

Rule 4



N States

0
1
⋮
N

M States

0
1
⋮
M

NM States

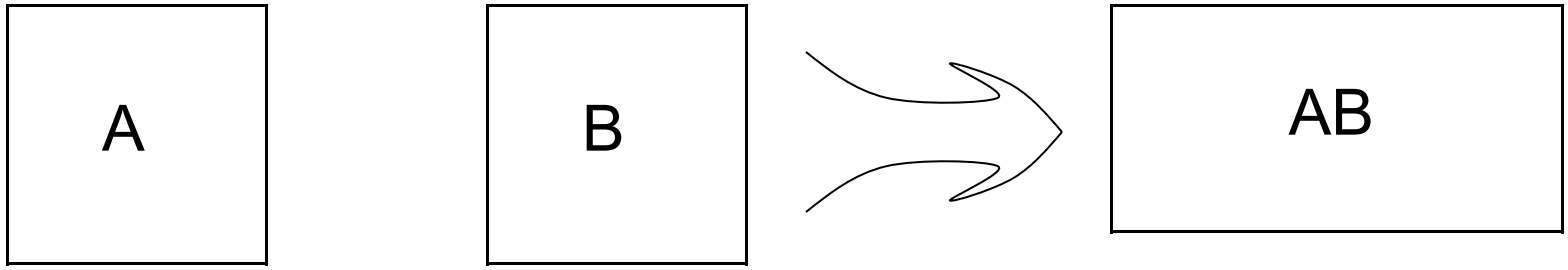
0,0	1,0	••	N,0
0,1	1,1		N,1
⋮	⋮		⋮
0,M	1,M	••	N,M

Probability vector in $\mathbb{R}^N \times \mathbb{R}^M$

$$\mathbb{R}^N \times \mathbb{R}^M = \mathbb{R}^{NM}$$

$$\mathbb{R}^N \times \mathbb{R}^M \neq \mathbb{R}^{N+M}$$

Rule 4 In Action

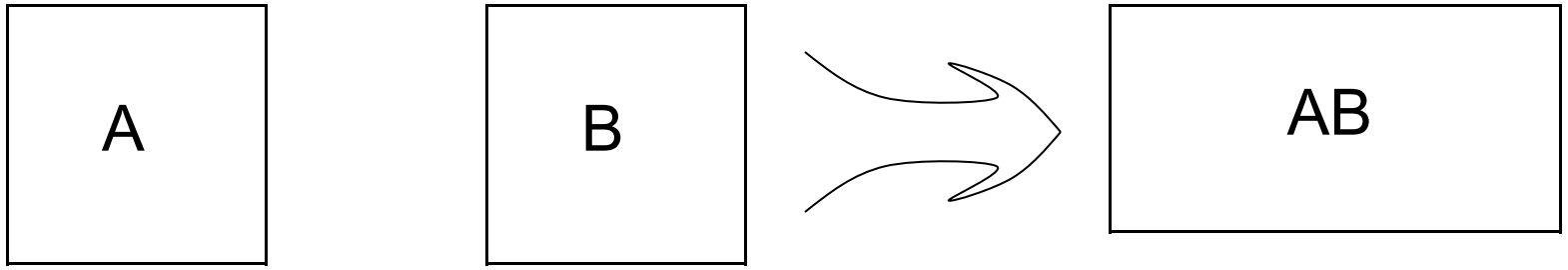


$$p = \begin{bmatrix} p_1 \\ 1-p_1 \end{bmatrix} \quad q = \begin{bmatrix} q_1 \\ 1-q_1 \end{bmatrix} \qquad r$$

$$r = p \otimes q = \begin{bmatrix} p_1 \\ 1-p_1 \end{bmatrix} \otimes \begin{bmatrix} q_1 \\ 1-q_1 \end{bmatrix} = \begin{bmatrix} p_1 q_1 \\ p_1(1-q_1) \\ (1-p_1)q_1 \\ (1-p_1)(1-q_1) \end{bmatrix} \begin{matrix} 00 \\ 01 \\ 10 \\ 11 \end{matrix}$$

Note that this is a description of a system which is not correlated. Each subsystem has its own probability vector.

Rule 4 In Action



contrast with $r = \begin{bmatrix} \frac{1}{2} \\ 0 \\ 0 \\ \frac{1}{2} \end{bmatrix} \begin{matrix} 00 \\ 01 \\ 10 \\ 11 \end{matrix}$.

It is impossible to express this probability vector as a tensor product:

$$r \neq p \otimes q$$

This is a description of a correlated system. Subsystem A is correlated with subsystem B.

Probabilistic Information Processing Device

Rule 1 (State Description) N states, probability vector

Rule 2 (Evolution) N x N stochastic matrix

Rule 3 (Measurement) k conditional probability vectors

Rule 4 (Composite Systems) tensor product

Quantum Information Processing Device

Rule 1 (State Description) N states, vector of amplitudes

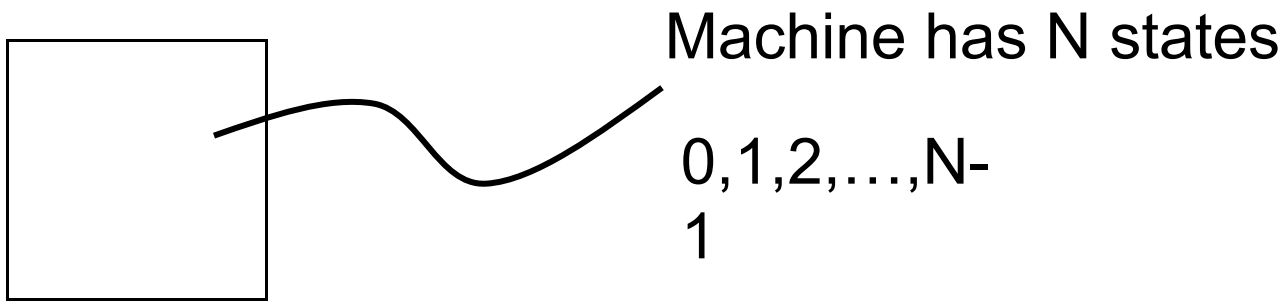
Rule 2 (Evolution) N x N unitary matrix

Rule 3 (Measurement) k measurement operators

Rule 4 (Composite Systems) tensor product

Quantum Rule 1

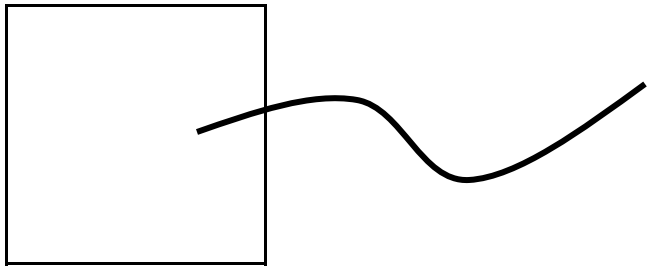
Rule 1 (State Description)



Rule 1 (State Description)

A quantum information processing machine is a machine with a state labeled from a finite alphabet of size N. Our description of the state of this system is a N dimensional **complex unit** vector

Quantum Rule 1



Machine has N states

0,1,2,…,N-1

N dimensional complex vector (vector of amplitudes)

$$v \in \mathbb{C}^N \qquad v = \begin{bmatrix} v_0 \\ v_1 \\ \vdots \\ v_{N-1} \end{bmatrix} \qquad v_i \in \mathbb{C}$$

Called: wave function, vector of amplitudes, quantum state

Complex Numbers

Complex numbers are numbers of the form

$$\alpha = a + bi$$

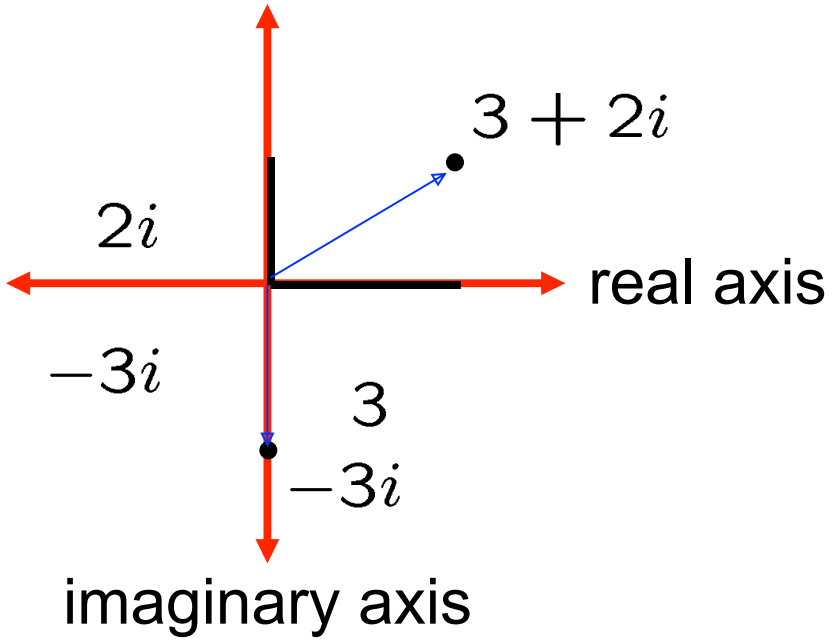
rea
l

rea
l

"square root of minus one"

$$i = \sqrt{-1} \qquad i^2 = -1$$

Complex plane:



Complex Numbers, Operations

Complex numbers can be added:

$$\alpha_1 = a_1 + b_1 i \qquad \alpha_2 = a_2 + b_2 i$$

$$\alpha_1 + \alpha_2 = (a_1 + a_2) + (b_1 + b_2)i$$

Complex numbers can be multiplied:

$$\alpha_1 = a_1 + b_1 i \qquad \alpha_2 = a_2 + b_2 i$$

$$\alpha_1\alpha_2 = (a_1+b_1 i)(a_2+b_2 i) = a_1a_2+a_1b_2 i+a_2b_1 i+b_1b_2 i^2$$
$$= (a_1a_2 - b_1b_2) + (a_1b_2 + a_2b_1)i$$

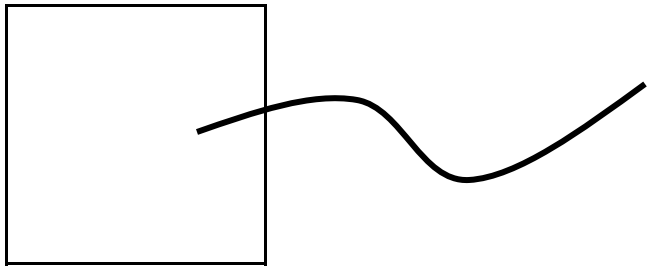
Complex numbers can be conjugated:

$$\alpha = a + bi$$
$$\alpha^* = a - bi$$

Modulus of a complex number

$$|\alpha|^2 = \alpha^*\alpha = a^2 + b^2$$

Quantum Rule 1



Machine has N states

0,1,2,…,N-1

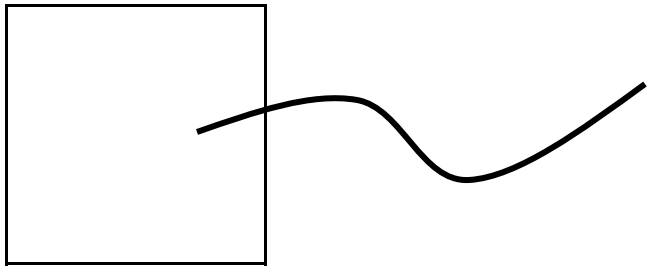
N dimensional complex vector (vector of amplitudes)

$$v \in \mathbb{C}^N \qquad v = \begin{bmatrix} v_0 \\ v_1 \\ \vdots \\ v_{N-1} \end{bmatrix} \qquad v_i \in \mathbb{C}$$

Complex unit vector:

$$\sum_{i=0}^{N-1} v_i^* v_i = \sum_{i=0}^{N-1} |v_i|^2 = v_0^* v_0 + v_1^* v_1 + \cdots + v_{N-1}^* v_{N-1} = 1$$

Quantum Rule 1



Machine has N states

0,1,2,…,N-1

N dimensional complex vector (vector of amplitudes)

$$v \in \mathbb{C}^N \qquad v = \begin{bmatrix} v_0 \\ v_1 \\ \vdots \\ v_{N-1} \end{bmatrix} \qquad v_i \in \mathbb{C}$$

Example: 2 state device

$$v = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} \end{bmatrix}$$

$$\sum_{i=0}^{N-1} |v_i|^2 = \frac{1}{\sqrt{2}}\frac{1}{\sqrt{2}} + \frac{-i}{\sqrt{2}}\frac{i}{\sqrt{2}} = 1$$

Quantum Information Processing Device

Rule 1 (State Description) N states, vector of amplitudes

Rule 2 (Evolution) N x N unitary matrix

The evolution in time of our description of the device is specified by an N x N unitary matrix U, such that if the description of the state before the evolution is given by the wave function v then the description of the system after this evolution is given by the wave function Uv

Unitary Matrix?

Unitary N x N matrix: an invertible N x N complex matrix whose inverse is equal to it's conjugate transpose.

Invertible: there exists an inverse of U, such that

$$U^{-1}U = I$$

N x N identity matrix

$$\begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & & \vdots \\ \vdots & & \ddots & \\ 0 & \cdots & & 1 \end{bmatrix}$$

$$\begin{bmatrix} U_{0,0} & \cdots & U_{0,N-1} \\ \vdots & & \vdots \\ U_{N-1,0} & \cdots & U_{N-1,N-1} \end{bmatrix}^{\dagger} = \begin{bmatrix} U^*_{0,0} & \cdots & U^*_{N-1,0} \\ \vdots & & \vdots \\ U^*_{0,N-1} & \cdots & U^*_{N-1,N-1} \end{bmatrix}$$

Unitary:

$$U^{-1} = U^{\dagger} \quad \text{or} \quad U^{\dagger}U = I \quad \text{or} \quad UU^{\dagger} = I$$

Quantum Rule 2, Example

$$U = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{-i}{\sqrt{2}} & \frac{i}{\sqrt{2}} \end{bmatrix}$$

Conjugate:

$$U^* = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} & \frac{-i}{\sqrt{2}} \end{bmatrix}$$

Conjugate transpose:

$$U^\dagger = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{-i}{\sqrt{2}} \end{bmatrix}$$

Unitary?

$$UU^\dagger = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{i}{\sqrt{2}} & \frac{i}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{-i}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$v = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

evolves to

$$v' = Uv = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{-i}{\sqrt{2}} & \frac{i}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Quantum Information Processing Device

Rule 1 (State Description) N states, vector of amplitudes

Rule 2 (Evolution) N x N unitary matrix

Rule 3 (Measurement) k measurement operators

Measurements in the computational basis of our device whose description is the wave function v results in one of N outcomes. The probability of the outcome i is given by the modulus of the ith component of v squared. After this measurement, the new state is given by a wave function with a single nonzero component: only the ith component is one.

Quantum Rule 1, Probabilities?

If we measure our quantum information processing machine, (in the computational basis) when our description is v, then the probability of observing state i is $|v_i|^2$

$$v = \begin{bmatrix} v_0 \\ v_1 \\ \vdots \\ v_{N-1} \end{bmatrix} \qquad \begin{bmatrix} |v_0|^2 \\ |v_1|^2 \\ \vdots \\ |v_{N-1}|^2 \end{bmatrix} \begin{matrix} Pr(0) \\ Pr(1) \\ \\ Pr(N-1) \end{matrix}$$

quantum state

probabilities

requirement of unit vector insures these are probabilities

Example:

$$v = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} \end{bmatrix}$$

$$Pr(0) = \frac{1}{\sqrt{2}}\frac{1}{\sqrt{2}} = \frac{1}{2} \qquad v' = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$Pr(1) = \frac{-i}{\sqrt{2}}\frac{i}{\sqrt{2}} = \frac{1}{2} \qquad v' = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Quantum Information Processing Device

Rule 1 (State Description) N states, vector of amplitudes

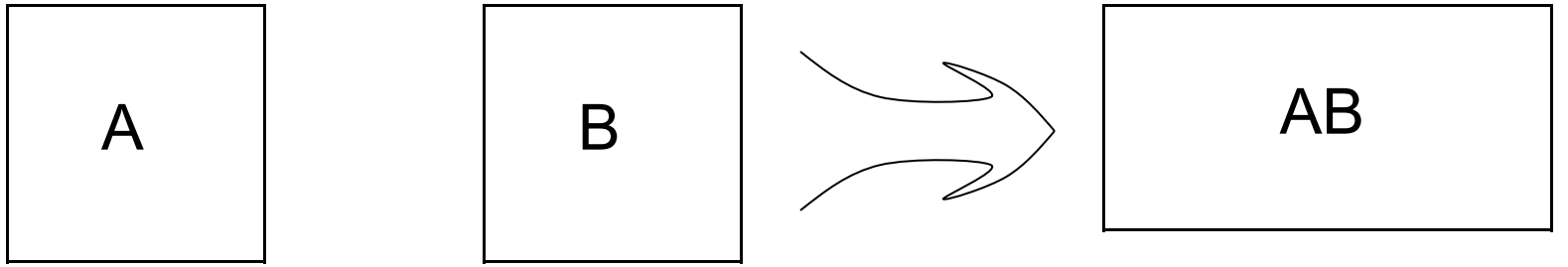
Rule 2 (Evolution) N x N unitary matrix

Rule 3 (Measurement) k measurement operators

Rule 4 (Composite Systems) tensor product

When combining two quantum systems with description spaces $\mathbb{C}^N$ and $\mathbb{C}^M$, the joint system is described by a Hilbert space which is a tensor product of these two systems, $\mathbb{C}^{NM} = \mathbb{C}^N \otimes \mathbb{C}^N$.

Quantum Rule 4



$\mathbb{C}^N$ $\mathbb{C}^M$ $\mathbb{C}^{NM} = \mathbb{C}^N \otimes \mathbb{C}^N$

0 0 $0 \otimes 0 = 00$

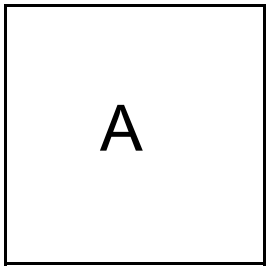
1 1 $0 \otimes 1 = 01$

N M $1 \otimes 0 = 10$

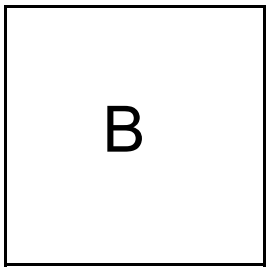
$N \otimes M = NM$

$dim(\mathcal{H}_A) = N$ $dim(\mathcal{H}_B) = M$ $dim(\mathcal{H}_{AB}) = NM$

Quantum Rule 4



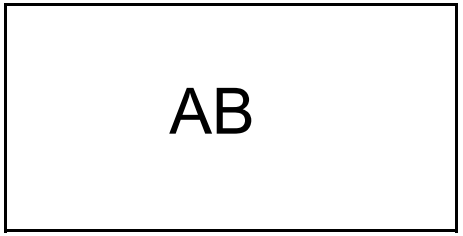
Example :



$$\psi_A = \begin{bmatrix} \alpha_A \\ \beta_A \end{bmatrix}$$



$$\psi_B = \begin{bmatrix} \alpha_B \\ \beta_B \end{bmatrix}$$



$$\psi_{AB} = \begin{bmatrix} \alpha_A \\ \beta_A \end{bmatrix} \otimes \begin{bmatrix} \alpha_B \\ \beta_B \end{bmatrix} = \begin{bmatrix} \alpha_A\alpha_B \\ \alpha_A\beta_B \\ \beta_A\alpha_A \\ \beta_A\beta_B \end{bmatrix}$$

separable state

Entangled States

Some joint states of two systems cannot be expressed as

$$\psi_{AB} = \psi_A \otimes \psi_B$$

Such states are called *entangled states*

Example :

$$\psi_{AB} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

We encountered something similar for our probabilistic device:

$$r = \begin{bmatrix} \frac{1}{2} \\ 0 \\ 0 \\ \frac{1}{2} \end{bmatrix} \begin{matrix} 00 \\ 01 \\ 10 \\ 11 \end{matrix} .$$

Entangled states are, similarly correlated.

But, we will find out later that they are correlated in a very peculiar manner!

Quantum Information Processing Device

Rule 1 (State Description) N states, vector of amplitudes

Rule 2 (Evolution) N x N unitary matrix

Rule 3 (Measurement) k measurement operators

Rule 4 (Composite Systems) tensor product

The Basic Postulates of Quantum Theory

These are the Laws

We need to learn notation, and all about quantum circuits.

Dirac Notation



David Mermin

"Mathematicians tend to despise Dirac notation, because it can prevent them from making important distinctions, but physicists love it, because they are always forgetting such distinctions exist and the notation liberates them from having to remember."

Examples of Dirac Notation:

$$|v\rangle \qquad \langle w|v\rangle \qquad |\langle w|U|v\rangle|^2 \qquad |v\rangle = \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle$$

$$|v\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \qquad U_{QFT} = \frac{1}{\sqrt{N}} \sum_{y=0}^{N-1} \sum_{x=0}^{N-1} \omega_N^{xy} |y\rangle\langle x|$$

Bras and Kets

$$|v\rangle = \begin{bmatrix} v_0 \\ v_1 \\ \vdots \\ v_{N-1} \end{bmatrix}$$

"ket" := column vector

$$\langle w| = \begin{bmatrix} w_0 & w_1 & \cdots w_{N-1} \end{bmatrix}$$

"bra" := row vector

Every ket has a unique bra obtained by complex conjugating and transposing:

$$|v\rangle = \begin{bmatrix} v_0 \\ v_1 \\ \vdots \\ v_{N-1} \end{bmatrix} \Leftrightarrow (|v\rangle)^\dagger = \langle v| = \begin{bmatrix} v_0^* & v_1^* & \cdots v_{N-1}^* \end{bmatrix}$$

Complex Vectors, Addition

Complex vectors can be added

$$|v\rangle = \begin{bmatrix} v_0 \\ v_1 \\ \vdots \\ v_{N-1} \end{bmatrix} \quad |w\rangle = \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_{N-1} \end{bmatrix}$$

$$|v\rangle + |w\rangle = \begin{bmatrix} v_0 + w_0 \\ v_1 + w_1 \\ \vdots \\ v_{N-1} + w_{N-1} \end{bmatrix}$$

Addition and multiplication by a scalar:

$$\alpha|v\rangle + \beta|w\rangle = \begin{bmatrix} \alpha v_0 + \beta w_0 \\ \alpha v_1 + \beta w_1 \\ \vdots \\ \alpha v_{N-1} + \beta w_{N-1} \end{bmatrix}$$

Computational Basis

Some special vectors:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix} \quad \cdots \quad |N-1\rangle = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$

Example:

2 dimensional complex vectors (also known as: a qubit!)

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \qquad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Computational Basis

Vectors can be “expanded” in the computational basis:

$$|v\rangle = \begin{bmatrix} v_0 \\ v_1 \\ \vdots \\ v_{N-1} \end{bmatrix} = v_0 \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} + v_1 \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix} + v_{N-1} \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$

$$= v_0|0\rangle + v_1|1\rangle + \cdots + v_{N-1}|N-1\rangle$$

Example:

$$|v\rangle = \begin{bmatrix} 1+2i \\ 3 \end{bmatrix} = (1+2i) \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 3 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$= (1+2i)|0\rangle + 3|1\rangle$$

Computational Bras

Computational Basis, but now for bras:

$$\langle 0| = \begin{bmatrix} 1 & 0 & \cdots 0 \end{bmatrix}$$

$$\langle 1| = \begin{bmatrix} 0 & 1 & \cdots 0 \end{bmatrix}$$

$$\vdots$$

$$\langle N-1| = \begin{bmatrix} 0 & 0 & \cdots 1 \end{bmatrix}$$

$$\langle v| = \begin{bmatrix} v_0^* & v_1^* & \cdots v_{N-1}^* \end{bmatrix} = v_0^*\langle 0| + v_1^*\langle 1| + \cdots + v_{N-1}^*\langle N-1|$$

Example:

$$\langle v| = \begin{bmatrix} 2 & 3+2i \end{bmatrix} = 2\langle 0| + (3+2i)\langle 1|$$

The Inner Product

Given a “bra” and a “ket” we can calculate an “inner product”

$$\langle w| = \begin{bmatrix} w_0^* & w_1^* & \cdots w_{N-1}^* \end{bmatrix} \qquad |v\rangle = \begin{bmatrix} v_0 \\ v_1 \\ \vdots \\ v_{N-1} \end{bmatrix}$$

$$\langle w|v\rangle = \begin{bmatrix} w_0^* & w_1^* & \cdots w_{N-1}^* \end{bmatrix} \begin{bmatrix} v_0 \\ v_1 \\ \vdots \\ v_{N-1} \end{bmatrix}$$

$$= w_0^* v_0 + w_1^* v_1 + \cdots + w_{N-1}^* v_{N-1}$$

This is a generalization of the dot product for real vectors

The result of taking an inner product is a complex number

The Inner Product

$$\langle w|v\rangle = w_0^* v_0 + w_1^* v_1 + \cdots + w_{N-1}^* v_{N-1}$$

Example:

$$|v\rangle = \begin{bmatrix} 1 \\ 1+2i \end{bmatrix} \qquad |w\rangle = \begin{bmatrix} 3i \\ 3 \end{bmatrix}$$

$$\langle w| = \begin{bmatrix} -3i & 3 \end{bmatrix}$$

$$\langle w|v\rangle = \begin{bmatrix} -3i & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1+2i \end{bmatrix} = (-3i)\cdot 1 + 3(1+2i) = 3+3i$$

$$\langle v|w\rangle = \begin{bmatrix} 1 & 1-2i \end{bmatrix} \begin{bmatrix} 3i \\ 3 \end{bmatrix} = 1\cdot(3i) + (1-2i)3 = 3-3i$$

Complex conjugate of inner product: $(\langle w|v\rangle)^* = \langle v|w\rangle$

The Inner Product in Comp. Basis

$$\langle w|v\rangle = w_0^* v_0 + w_1^* v_1 + \cdots + w_{N-1}^* v_{N-1}$$

$$\langle 0|0\rangle = \begin{bmatrix} 1 & 0 & \cdots 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = 1{\cdot}1 + 0{\cdot}0 + \cdots + 0{\cdot}0 = 1$$

$$\langle 0|1\rangle = \begin{bmatrix} 1 & 0 & \cdots 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix} = 1{\cdot}0 + 1{\cdot}0 + \cdots + 0{\cdot}0 = 0$$

Inner product of computational basis elements:

$$\langle j|k\rangle = \delta_{j,k} = \begin{cases} 1 \text{ if } j = k \\ 0 \text{ otherwise} \end{cases}$$

Kronecker delta

The Inner Product in Comp. Basis

$$\langle w|v\rangle = w_0^* v_0 + w_1^* v_1 + \cdots + w_{N-1}^* v_{N-1}$$

$$\langle w| = w_0^*\langle 0| + w_1^*\langle 1| + \cdots + w_{N-1}^*\langle N-1|$$

$$|v\rangle = v_0|0\rangle + v_1|1\rangle + \cdots + v_{N-1}|N-1\rangle$$

$$\begin{aligned}\langle w|v\rangle \;=\; & (w_0^*\langle 0| + w_1^*\langle 1| + \cdots + w_{N-1}^*\langle N-1|) \\ & (v_0|0\rangle + v_1|1\rangle + \cdots + v_{N-1}|N-1\rangle)\end{aligned}$$

$$\langle w|v\rangle = w_0^* v_0 + w_1^* v_1 + \cdots + w_{N-1}^* v_{N-1}$$

Example $|v\rangle = |0\rangle + 2i|1\rangle$ $|w\rangle = 3i|0\rangle + (2i+2)|1\rangle$

:

$$\langle w|v\rangle = -3i \cdot 1 + (-2i+2)2i = 4 + i$$

Norm of a Vector

Norm of a vector

$$|||v\rangle|| = \sqrt{\langle v|v\rangle}$$

$$\langle v|v\rangle = v_0^* v_0 + v_1^* v_1 + \cdots + v_{N-1}^* v_{N-1}$$
$$= |v_0|^2 + |v_1|^2 + \cdots + |v_{N-1}|^2$$

which is always a positive real number

it is the length of the complex vector

Example $|v\rangle = |0\rangle + 2i|1\rangle$

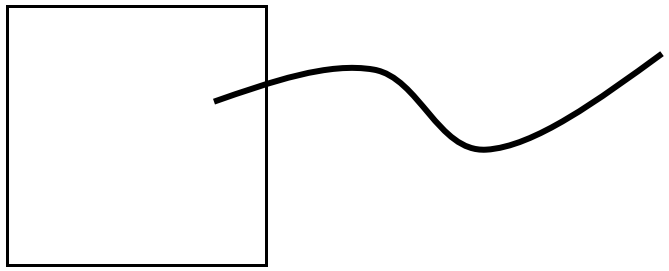
:

$$\langle v|v\rangle = |1|^2 + |2i|^2 = 5$$

$$|||v\rangle|| = \sqrt{5}$$

Quantum Rule 1

Rule 1 (State Description)



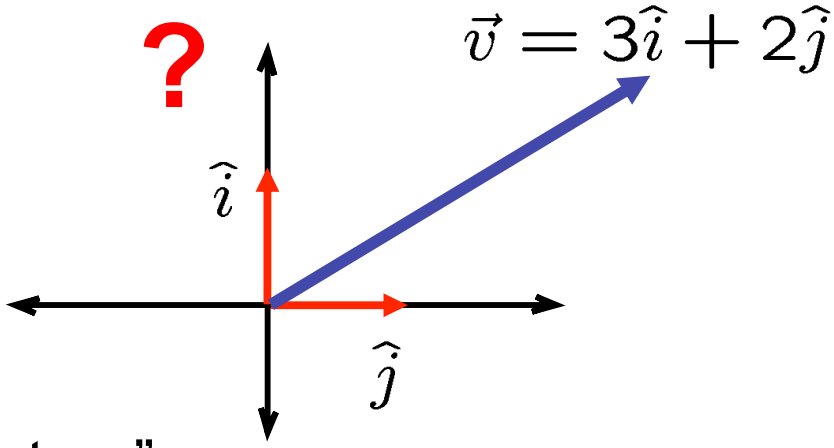
Rule 1 (State Description)

A quantum information processing machine is a machine with a state labeled from a finite alphabet of size N. Our description of the state of this system is a N dimensional **complex unit** vector

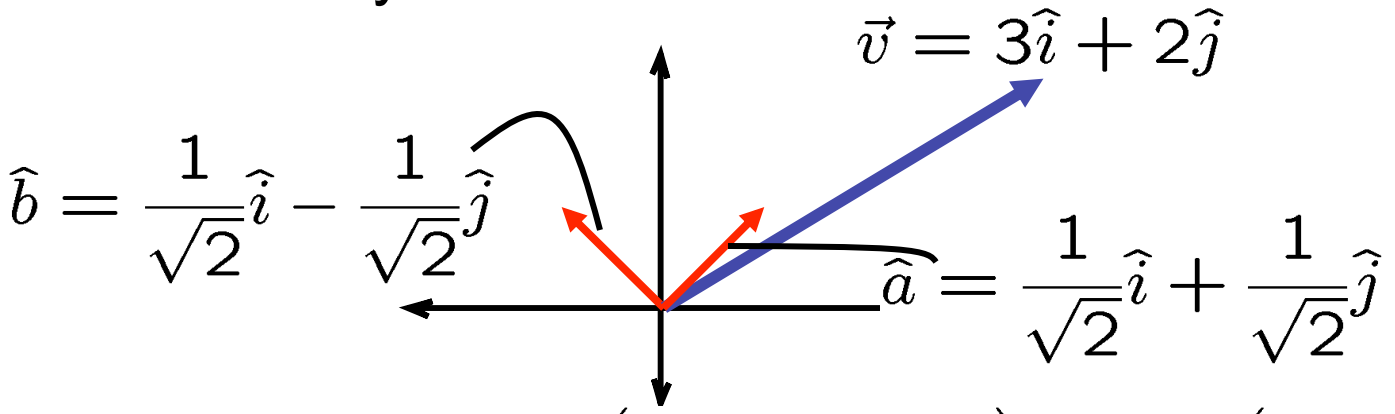
$$|v\rangle = v_0|0\rangle + v_1|1\rangle + \cdots + v_{N-1}|N-1\rangle$$

$$\langle v|v\rangle = 1 \quad \textit{normalized}$$

Basis ?



"Other coordinate system"



$$\vec{v} = \frac{5}{\sqrt{2}}\hat{a} + \frac{1}{\sqrt{2}}\hat{b} = \frac{5}{\sqrt{2}}\left(\frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j}\right) + \frac{1}{\sqrt{2}}\left(\frac{1}{\sqrt{2}}\hat{i} - \frac{1}{\sqrt{2}}\hat{j}\right)$$

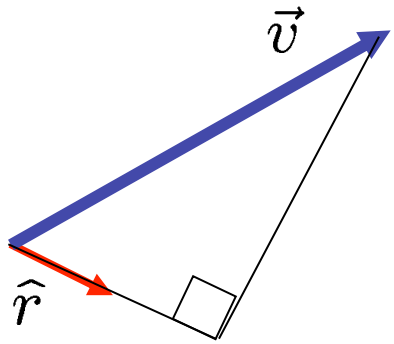
$$= \left(\frac{5}{2} + \frac{1}{2}\right)\hat{i} + \left(\frac{5}{2} - \frac{1}{2}\right)\hat{j}$$

Resolving a Vector

use the dot product to get the component of a vector
along a direction:

unit vector

$$|\hat{r}| = 1$$



$$\hat{r} \cdot \vec{v}$$

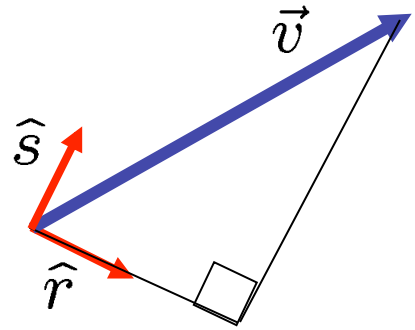
use two orthogonal unit vectors in 2D to write in new basis:

orthogonal unit vectors:

$$|\hat{r}| = 1$$

$$|\hat{s}| = 1$$

$$\hat{r} \cdot \hat{s} = 0$$

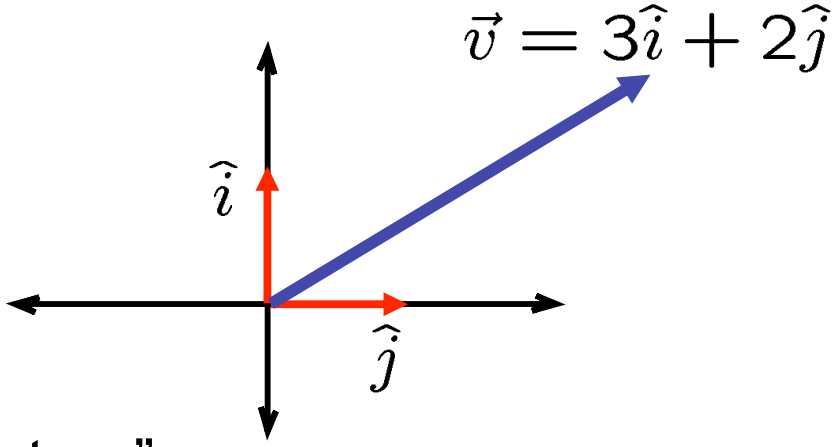


$$\hat{s} \cdot \vec{v}$$

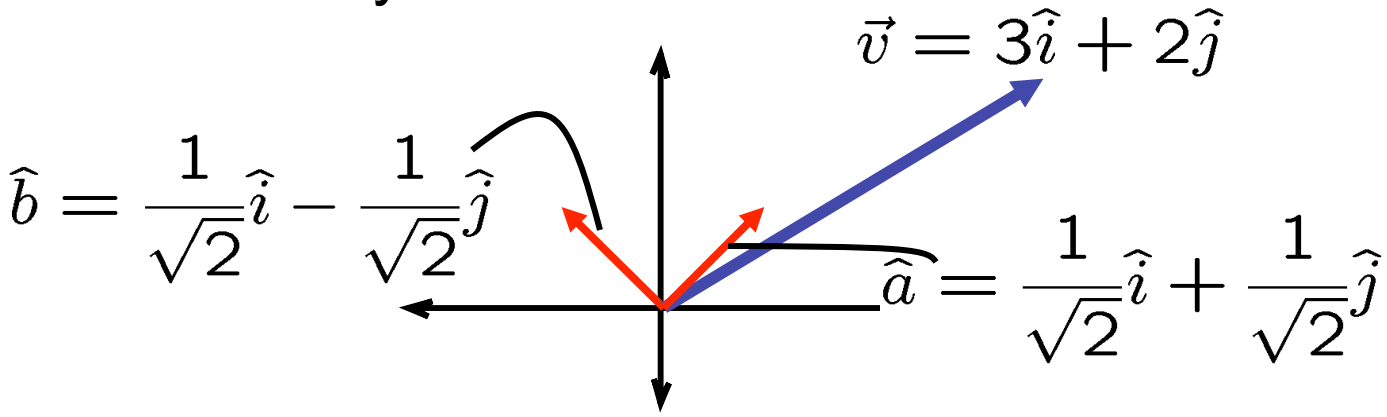
$$\vec{v} = (\hat{r} \cdot \vec{v})\hat{r} + (\hat{s} \cdot \vec{v})\hat{s}$$

$$\hat{r} \cdot \vec{v}$$

Expressing In a New Basis



"Other coordinate system"



$$\hat{a} \cdot \vec{v} = 3(\frac{1}{\sqrt{2}}) + 2(\frac{1}{\sqrt{2}}) = \frac{5}{\sqrt{2}}$$

$$\hat{b} \cdot \vec{v} = 3(\frac{1}{\sqrt{2}}) - 2(\frac{1}{\sqrt{2}}) = \frac{1}{\sqrt{2}}$$

$$\vec{v} = \frac{5}{\sqrt{2}}\hat{a} + \frac{1}{\sqrt{2}}\hat{b}$$

Computational Basis

Computational basis:

$$|0\rangle$$
$$|1\rangle$$
$$\vdots$$
$$|N-1\rangle$$

is an orthonormal basis:

$$\langle j|k\rangle = \delta_{j,k} = \begin{cases} 1 \text{ if } j = k \\ 0 \text{ otherwise} \end{cases}$$

Kronecker delta

Computational basis is important because when we measure our quantum computer (a qubit, two qubits, etc.) we get an outcome corresponding to these basis vectors.

But there are all sorts of other basis which we could use to, say, expand our vector about.

A Different Basis

A different orthonormal basis:

$$|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

$$|-\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$$

$$\langle +|+\rangle = \left|\frac{1}{\sqrt{2}}\right|^2 + \left|\frac{1}{\sqrt{2}}\right|^2 = 1$$

$$\langle -|-\rangle = \left|\frac{1}{\sqrt{2}}\right|^2 + \left|\frac{-1}{\sqrt{2}}\right|^2 = 1$$

$$\langle +|-\rangle = \left(\frac{1}{\sqrt{2}}\frac{1}{\sqrt{2}}\right) + \left(\frac{1}{\sqrt{2}}\frac{-1}{\sqrt{2}}\right) = 0$$

An orthonormal basis is complete if the number of basis elements is equal to the dimension of the complex vector space.

Changing Your Basis

Express the qubit wave function $|v\rangle = v_0|0\rangle + v_1|1\rangle$ in the orthonormal complete basis

$$|a\rangle, |b\rangle$$

in other words find components of $|v\rangle = v_a|a\rangle + v_b|b\rangle$

Some inner products:

$$\langle a|v\rangle = \langle a|\left(v_a|a\rangle + v_b|b\rangle\right) = v_a\langle a|a\rangle + v_b\langle a|b\rangle = v_a$$

$$\langle b|v\rangle = \langle b|\left(v_a|a\rangle + v_b|b\rangle\right) = v_a\langle b|a\rangle + v_b\langle b|b\rangle = v_b$$

So:

$$|v\rangle = (\langle a|v\rangle)|a\rangle + (\langle b|v\rangle)|b\rangle$$

Calculating these inner products allows us to express the ket in a new basis.

Example Basis Change

$$|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle \qquad |-\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$$

Express $|v\rangle = v_0|0\rangle + v_1|1\rangle$ in this basis: $|v\rangle = v_+|+\rangle + v_-|-\rangle$

$$\langle +|v\rangle = \langle +|\left(v_+|+\rangle + v_-|-\rangle\right) = v_+\langle +|+\rangle + v_-\langle +|-\rangle = v_+$$

$$\langle -|v\rangle = \langle -|\left(v_+|+\rangle + v_-|-\rangle\right) = v_+\langle -|+\rangle + v_-\langle -|-\rangle = v_-$$

So $|v\rangle = (\langle +|v\rangle)|+\rangle + (\langle -|v\rangle)|-\rangle$:

$$\langle +|v\rangle = \left(\frac{1}{\sqrt{2}}v_0\right) + \left(\frac{1}{\sqrt{2}}v_1\right) = \frac{v_0 + v_1}{\sqrt{2}}$$

$$\langle -|v\rangle = \left(\frac{1}{\sqrt{2}}v_0\right) + \left(\frac{-1}{\sqrt{2}}v_1\right) = \frac{v_0 - v_1}{\sqrt{2}}$$

$$|v\rangle = \frac{v_0 + v_1}{\sqrt{2}}|+\rangle + \frac{v_0 - v_1}{\sqrt{2}}|-\rangle$$

Explicit Basis Change

$$|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle \qquad |-\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$$

Express $|v\rangle = |0\rangle$ in this basis: $|v\rangle = v_+|+\rangle + v_-|-\rangle$

$$\langle +|0\rangle = \frac{1}{\sqrt{2}}$$

$$\langle -|0\rangle = \frac{1}{\sqrt{2}}$$

$$|v\rangle = (\langle +|v\rangle)|+\rangle + (\langle -|v\rangle)|-\rangle$$

So:

$$|v\rangle = \frac{1}{\sqrt{2}}|+\rangle + \frac{1}{\sqrt{2}}|-\rangle$$

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} + \frac{1}{\sqrt{2}} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} + \frac{1}{2} \\ \frac{1}{2} - \frac{1}{2} \end{bmatrix}$$

Linear Algebra

Linear algebra is the language of quantum theory.

We will need to develop some basic proficiency in order to work with quantum computers.

Matrices

A N dimensional complex matrix M is an N by N array of complex numbers:

$$M = \begin{bmatrix} M_{0,0} & \cdots & M_{0,N-1} \\ \vdots & & \vdots \\ M_{N-1,0} & \cdots & M_{N-1,N-1} \end{bmatrix}$$

$M_{j,k}$ are complex numbers

Example

- Three dimensional complex matrix:

$$M = \begin{bmatrix} 4 & 3+i & 2 \\ i & e^{\frac{\pi}{4}} & \sqrt{2}i \\ 0 & 0 & 4 \end{bmatrix}$$

$$M_{1,0} = i$$

$$M_{2,2} = 4$$

Matrices, Multiplied by Scalar

Matrices can be multiplied by a complex number

$$M = \begin{bmatrix} M_{0,0} & \cdots & M_{0,N-1} \\ \vdots & & \vdots \\ M_{N-1,0} & \cdots & M_{N-1,N-1} \end{bmatrix}$$

$$\alpha M = \begin{bmatrix} \alpha M_{0,0} & \cdots & \alpha M_{0,N-1} \\ \vdots & & \vdots \\ \alpha M_{N-1,0} & \cdots & \alpha M_{N-1,N-1} \end{bmatrix}$$

Example:

$$M = \begin{bmatrix} 0 & 3+i \\ i & 1 \end{bmatrix} \qquad \alpha = 2i$$

$$\alpha M = \begin{bmatrix} 2i \cdot 0 & 2i(3+i) \\ 2i(i) & 2i(1) \end{bmatrix} = \begin{bmatrix} 0 & -2+6i \\ -2 & 2i \end{bmatrix}$$

Matrices, Added

Matrices can be added

$$M = \begin{bmatrix} M_{0,0} & \cdots & M_{0,N-1} \\ \vdots & & \vdots \\ M_{N-1,0} & \cdots & M_{N-1,N-1} \end{bmatrix} L = \begin{bmatrix} L_{0,0} & \cdots & L_{0,N-1} \\ \vdots & & \vdots \\ L_{N-1,0} & \cdots & L_{N-1,N-1} \end{bmatrix}$$

$$M+L = \begin{bmatrix} M_{0,0}+L_{0,0} & \cdots & M_{0,N-1}+L_{0,N-1} \\ \vdots & & \vdots \\ M_{N-1,0}+L_{N-1,0} & \cdots & M_{N-1,N-1}+L_{N-1,N-1} \end{bmatrix}$$

Example:

$$M = \begin{bmatrix} 0 & 3+i \\ i & 1 \end{bmatrix} \qquad L = \begin{bmatrix} 3 & -3-i \\ i & 2 \end{bmatrix}$$

$$M+L = \begin{bmatrix} 0+3 & 3+i+(-3-i) \\ i+i & 1+2 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 2i & 3 \end{bmatrix}$$

Matrices, Multiplied

Matrices can be multiplied

$$M = \begin{bmatrix} M_{0,0} & \cdots & M_{0,N-1} \\ \vdots & & \vdots \\ M_{N-1,0} & \cdots & M_{N-1,N-1} \end{bmatrix} L = \begin{bmatrix} L_{0,0} & \cdots & L_{0,N-1} \\ \vdots & & \vdots \\ L_{N-1,0} & \cdots & L_{N-1,N-1} \end{bmatrix}$$

$$R = ML = \begin{bmatrix} R_{0,0} & \cdots & R_{0,N-1} \\ \vdots & & \vdots \\ R_{N-1,0} & \cdots & R_{N-1,N-1} \end{bmatrix}$$

$$R_{0,0} = M_{0,0}L_{0,0} + M_{0,1}L_{1,0} + \cdots + M_{0,N-1}L_{N-1,0}$$
$$R_{0,1} = M_{0,0}L_{0,1} + M_{0,1}L_{1,1} + \cdots + M_{0,N-1}L_{N-1,1}$$
$$\vdots$$
$$R_{j,k} = M_{j,0}L_{0,k} + M_{j,1}L_{1,k} + \cdots + M_{j,N-1}L_{N-1,k}$$

$$R_{j,k} = \sum_{l=0}^{N-1} M_{j,l}L_{l,k}$$

Matrices and Kets, Multiplied

Given a matrix, and a column vector:

$$M = \begin{bmatrix} M_{0,0} & \cdots & M_{0,N-1} \\ \vdots & & \vdots \\ M_{N-1,0} & \cdots & M_{N-1,N-1} \end{bmatrix} \qquad |v\rangle = \begin{bmatrix} v_0 \\ v_1 \\ \vdots \\ v_{N-1} \end{bmatrix}$$

These can be multiplied to obtain a new column vector:

$$M|v\rangle = \begin{bmatrix} M_{0,0} & \cdots & M_{0,N-1} \\ \vdots & & \vdots \\ M_{N-1,0} & \cdots & M_{N-1,N-1} \end{bmatrix} \begin{bmatrix} v_0 \\ \vdots \\ v_{N-1} \end{bmatrix}$$

$$= \begin{bmatrix} M_{0,0}v_0 + M_{0,1}v_1 + \cdots M_{0,N-1}v_{N-1} \\ \vdots \\ M_{N-1,0}v_0 + M_{N-1,1}v_1 + \cdots M_{N-1,N-1}v_{N-1} \end{bmatrix}$$

Matrices and Bras, Multiplied

Given a matrix, and a row vector:

$$M = \begin{bmatrix} M_{0,0} & \cdots & M_{0,N-1} \\ \vdots & & \vdots \\ M_{N-1,0} & \cdots & M_{N-1,N-1} \end{bmatrix} \quad \langle w| = \begin{bmatrix} w_0^* & w_1^* & \cdots w_{N-1}^* \end{bmatrix}$$

These can be multiplied to obtain a new row vector:

$$\langle w|M = \begin{bmatrix} w_0^* & \cdots & w_{N-1}^* \end{bmatrix} \begin{bmatrix} M_{0,0} & \cdots & M_{0,N-1} \\ \vdots & & \vdots \\ M_{N-1,0} & \cdots & M_{N-1,N-1} \end{bmatrix}$$

$$\langle w|M = \begin{bmatrix} r_0^* & \cdots & r_{N-1}^* \end{bmatrix}$$

$$\begin{aligned} r_0^* &= w_0^* M_{0,0} + \cdots + w_{N-1}^* M_{N-1,0} \\ &\vdots \\ r_{N-1}^* &= w_0^* M_{0,N-1} + \cdots + w_{N-1}^* M_{N-1,N-1} \end{aligned}$$

Matrices, Complex Conjugate

Given a matrix, we can form its complex conjugate by conjugating every element:

$$M = \begin{bmatrix} M_{0,0} & \cdots & M_{0,N-1} \\ \vdots & & \vdots \\ M_{N-1,0} & \cdots & M_{N-1,N-1} \end{bmatrix}$$

$$M^* = \begin{bmatrix} M^*_{0,0} & \cdots & M^*_{0,N-1} \\ \vdots & & \vdots \\ M^*_{N-1,0} & \cdots & M^*_{N-1,N-1} \end{bmatrix}$$

Example:

$$M = \begin{bmatrix} 0 & 3+i \\ i & 1 \end{bmatrix} \qquad M^* = \begin{bmatrix} 0 & 3-i \\ -i & 1 \end{bmatrix}$$

Matrices, Transpose

Given a matrix, we can form it's transpose by reflecting across the diagonal

$$M = \begin{bmatrix} M_{0,0} & \cdots & M_{0,N-1} \\ \vdots & & \vdots \\ M_{N-1,0} & \cdots & M_{N-1,N-1} \end{bmatrix}$$

$$M^T = \begin{bmatrix} M_{0,0} & \cdots & M_{N-1,0} \\ \vdots & & \vdots \\ M_{0,N-1} & \cdots & M_{N-1,N-1} \end{bmatrix}$$

Example:

$$M = \begin{bmatrix} 0 & 3+i \\ i & 1 \end{bmatrix} \qquad M^T = \begin{bmatrix} 0 & i \\ 3+i & 1 \end{bmatrix}$$

Matrices, Conjugate Transpose

Given a matrix, we can form its conjugate transpose by reflecting across the diagonal and conjugating

$$M = \begin{bmatrix} M_{0,0} & \cdots & M_{0,N-1} \\ \vdots & & \vdots \\ M_{N-1,0} & \cdots & M_{N-1,N-1} \end{bmatrix}$$

$$M^\dagger = \begin{bmatrix} M^*_{0,0} & \cdots & M^*_{N-1,0} \\ \vdots & & \vdots \\ M^*_{0,N-1} & \cdots & M^*_{N-1,N-1} \end{bmatrix}$$

Example:

$$M = \begin{bmatrix} 0 & 3+i \\ i & 1 \end{bmatrix} \qquad M^\dagger = \begin{bmatrix} 0 & -i \\ 3-i & 1 \end{bmatrix}$$

Unitary Matrices

A matrix U is unitary if

$$U^\dagger U = I$$

N x N identity matrix

$$I = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & & \vdots \\ \vdots & & \ddots & \\ 0 & \cdots & & 1 \end{bmatrix} = diag(1, 1, \ldots, 1)$$

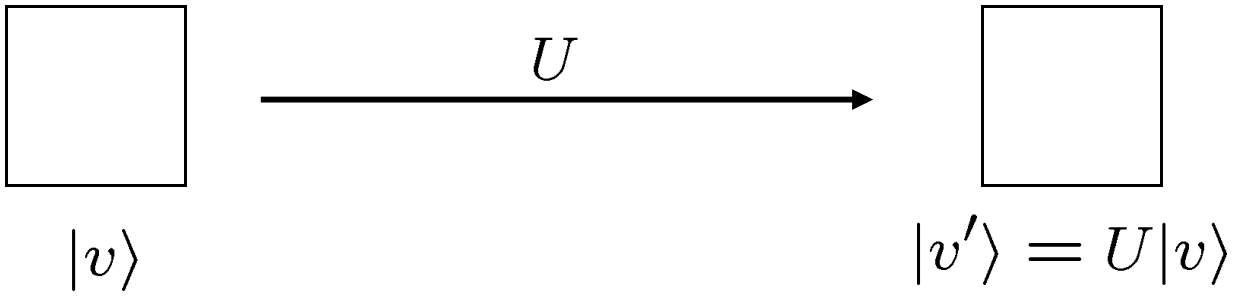
Equivalently a matrix U is unitary if

$$UU^\dagger = I$$

Quantum Rule 2

Rule 2: The wave function of a N dimensional quantum system evolves in time according to a unitary matrix U. If the wave function initially is $|v\rangle$ then after the evolution correspond to U the new wave function is $|v'\rangle = U|v\rangle$

"Unitary Evolution"



Unitary Evolution and the Norm

Unitary evolution

$$|v'\rangle = U|v\rangle$$

What happens to the norm $\langle v'|v'\rangle$ of the ket?

$$\langle v'| = (U|v\rangle)^\dagger = \langle v|U^\dagger$$

$$\langle v'|v'\rangle = \langle v|U^\dagger U|v\rangle = \langle v|I|v\rangle = \langle v|v\rangle$$

Unitary evolution does not change the length of the ket.

Normalized wave function

$$\sqrt{\langle v|v\rangle} = 1$$

Normalized wave function

$$|v'\rangle = U|v\rangle \xrightarrow{\text{unitary evolution}} \sqrt{\langle v'|v'\rangle} = 1$$

This implies that unitary evolution will maintain being a unit vector